

Computer Science & IT

Computer Networks

DPP: 1

Introduction

Q1 Which of the following is a function of the application layer ?

- (A) Packet routing and forwarding
- (B) Provide network services to users
- (C) Framing and error control
- (D) Host-to-host connectivity

Q2 Which layer of OSI model is responsible for data encryption and decryption ?

- (A) Application Layer
- (B) Presentation Layer
- (C) Session Layer
- (D) Transport Layer

Q3 Which layer in the OSI model is providing end-to-end connectivity ?

- (A) Application Layer
- (B) Transport Layer
- (C) Network Layer
- (D) Data Link Layer

Q4 The protocol data unit (PDU) for the transport layer in the Internet stack is :

- (A) Frame
- (B) Datagram
- (C) Segment
- (D) Message

Q5 Which of the following transport layer protocol uses for file transfer application ?

- (A) SMTP
- (B) FTP
- (C) TCP
- (D) UDP

Q6 Router is a _____ layer networking device.

- (A) Application Layer
- (B) Transport Layer

(C) Network Layer

(D) Data Link Layer

Q7 Which of the following is a function of the network layer ?

- (A) Data encryption and decryption
- (B) Error detection and correction
- (C) Routing and forwarding
- (D) Multiplexing and demultiplexing

Q8 Which of the following address type is used by the network layer ?

- (A) MAC Address
- (B) IP Address
- (C) Port Number
- (D) None of the above

Q9 Match the following :

List-I	List-II
P: Segment	I: Application Layer
Q: Frame	II: Transport Layer
R: Datagram	III: Network Layer
S: Message	IV: Data Link Layer

(A) P-II, Q-IV, R-I, S-III

(B) P-III, Q-IV, R-II, S-I

(C) P-II, Q-IV, R-III, S-I

(D) P-II, Q-III, R-IV, S-I

Q10 Assume that source S and destination D are connected through one router. Determine how many times each packet has to visit the network layer during transmission from S to D ?


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- Q11** How many maximum number of links in mesh topology with 6 nodes ?
- Q12** Consider that 12 machines need to be connected in a LAN using 5-port ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is _____ .
- Q13** _____ is an example of a circuit-switched network .
(A) Internet chat
- (B) Telephone network (PSTN - Public Switched Telephone Network)
- (C) Web browsing
- (D) Email
- Q14** Which switching techniques provide "connection-less" services ?
(A) Circuit Switching
(B) Packet Switching
(C) Virtual-circuit Switching
(D) None of the above



Answer Key

Q1 (B)

Q2 (B)

Q3 (B)

Q4 (C)

Q5 (C)

Q6 (C)

Q7 (C)

Q8 (B)

Q9 (C)

Q10 3~3

Q11 15~15

Q12 4~4

Q13 (B)

Q14 (B)



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Hints & Solutions

Q1 Text Solution:

Provide network services to users

Q2 Text Solution:

Presentation Layer

Q3 Text Solution:

Transport Layer → End to End (Process to Process)

Q4 Text Solution:

Segment → Transport Layer (PDU)

Q5 Text Solution:

TCP

FTP → TCP → IP

Q6 Text Solution:

Network Layer (Layer -3) (Router)

Q7 Text Solution:

Routing and forwarding → Network layer

Q8 Text Solution:

IP Address → Transport layer

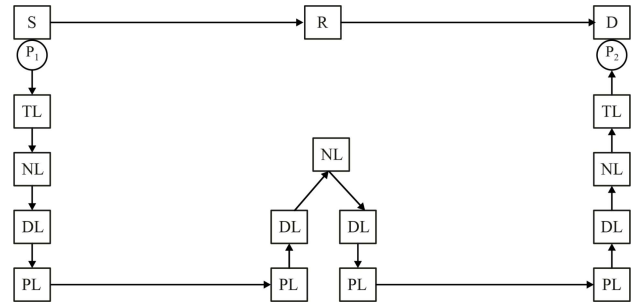
Q9 Text Solution:

P: Segment → II: Transport Layer

Q: Frame → IV: Data Link Layer

R: Datagram → III: Network Layer

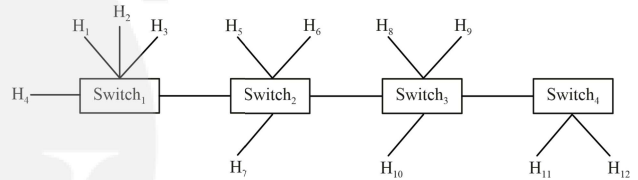
S: Message → I: Application Layer

Q10 Text Solution:**Q11 Text Solution:**

Fully mesh topology

Number of Nodes = $N = 6$

$$\text{Maximum number of Links} = \frac{N*(N-1)}{2} = N_{C_2} \\ = \frac{6*5}{2} = 15$$

Q12 Text Solution:**Q13 Text Solution:**

Telephone network (PSTN - Public Switched Telephone Network)

Q14 Text Solution:

Packet Switching → connection less and unreliable



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